TryHackMe Room Writeup 2 of 6

TryHackMe: Nmap Live Host Discovery

Network Traffic Analysis, Nmap and Windows Logging | Week 2 | Lab Part

Room detail	Details
Platform	TryHackMe
Room	Nmap Live Host Discovery
Room link	https://tryhackme.com/room/nmap01
Public profile	https://tryhackme.com/p/hariniedward
Status	Completed, 100%
About the room	Learn how to use Nmap to discover live hosts using ARP, ICMP, and TCP and UDP ping scans.

Objective

This is the first of four Nmap rooms, and it answers one question: which systems on a network are actually up? Finding live hosts first saves time and noise, because there is no reason to port scan addresses that nobody uses.

What I learned

- A network segment is a physical connection, while a subnet is a logical one with its own address range. A /24 holds about 250 hosts and a /16 about 65,000.
- ARP only works inside the local subnet, because ARP packets are never routed.
- Targets can be given to Nmap as a list, a range such as 10.11.12.15-20, a CIDR subnet, or a file with -iL. The -sL option lists the targets without scanning them.
- By default, a privileged user scanning a local network gets ARP requests. Outside the local network Nmap sends ICMP echo, a TCP ACK to port 80, a TCP SYN to port 443 and an ICMP timestamp request.
- Each layer offers a way to find a host: ARP at the link layer, ICMP at the network layer, and TCP or UDP at the transport layer.
- Reverse DNS lookups can reveal useful host names. The -R option forces them for every address and -n turns them off.
- Masscan works in a similar way but sends packets much faster.

Host discovery methods

Method	Option	What proves the host is up	Limitation
ARP scan	-PR -sn	An ARP reply carrying the host's MAC address.	Same subnet only.
ICMP echo	-PE -sn	An ICMP echo reply (type 0).	Often blocked, including by the default Windows firewall.
ICMP timestamp	-PP -sn	An ICMP timestamp reply (type 14).	Sometimes works when echo is blocked.
ICMP address mask	-PM -sn	An ICMP address mask reply (type 18).	Frequently blocked. It found nothing in the lab.
TCP SYN ping	-PS -sn	A SYN-ACK or RST to a SYN sent to a port, 80 by default.	Does not need root, but completes a handshake without it.
TCP ACK ping	-PA -sn	An RST in reply to an ACK that belongs to no connection.	Needs a privileged user.

Method	Option	What proves the host is up	Limitation
UDP ping	-PU -sn	An ICMP port unreachable reply from a closed UDP port.	An open UDP port stays silent.

The practical task

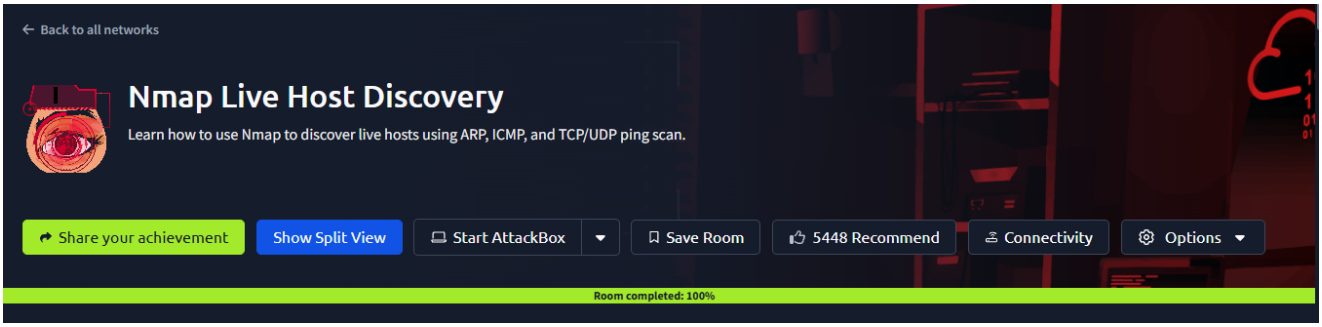
- Used the network simulator to send ARP requests and see which devices receive them, and why a computer on another subnet never answers.
- Watched how a ping on the same subnet is always preceded by an ARP request.
- Worked out how many addresses a given subnet or range makes Nmap scan.
- Ran the different ping scan types against the lab subnet from the AttackBox and compared how many hosts each one found.

Why this matters for a SOC analyst

- Host discovery leaves clear patterns: sequential ARP requests across a subnet, or ICMP and SYN packets sent to every address in a range.
- A sudden ARP sweep from a workstation can mean someone is mapping the internal network after a compromise.
- Because ICMP address mask found nothing while other methods worked, I understood why blocking one type of packet does not hide a host from a scanner.

> **Key takeaway**
> No single probe finds every host. Using ARP, ICMP, TCP and UDP together gives the full picture, and the same patterns are exactly what a defender should look for in the traffic.

Completion proof



Room page showing 100% completion. My profile details have been removed from the screenshot.

References

Room: https://tryhackme.com/room/nmap01

My public profile: https://tryhackme.com/p/hariniedward